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MA3521 — Monday/Tuesday Worksheet Key
Do not row reduce or solve.

Name:

1. $\mathbf{y}' = \boldsymbol{A}\mathbf{y} + e^{-2t}\begin{pmatrix}-3\\-5\\-1\end{pmatrix} \quad \boldsymbol{A} = \begin{pmatrix}1 & -2 & 0\\ 2 & 1 & 0\\ 0 & 0 & -1\end{pmatrix} \quad \mathbf{y}(0) = \begin{pmatrix}2\\3\\1\end{pmatrix}$

   Eigenpairs $1+2i,\ \begin{pmatrix}1\\-i\\0\end{pmatrix} \quad 1-2i,\ \begin{pmatrix}1\\i\\0\end{pmatrix} \quad -1,\ \begin{pmatrix}0\\0\\1\end{pmatrix}$

   (a) $\mathbf{y}_c(t) =$ $\begin{pmatrix}e^t\cos(2t) & -e^t\sin(2t) & 0\\ e^t\sin(2t) & e^t\cos(2t) & 0\\ 0 & 0 & e^{-t}\end{pmatrix}\mathbf{c}$

   $\alpha = 1, \quad \beta = 2$

   $\operatorname{re}\mathbf{v} = \begin{pmatrix}1\\0\\0\end{pmatrix}, \quad \operatorname{im}\mathbf{v} = \begin{pmatrix}0\\-1\\0\end{pmatrix}$

   $\operatorname{re}\left(e^{(1+2i)t}\mathbf{v}\right) = e^t\begin{pmatrix}\cos(2t)\\ \sin(2t)\\ 0\end{pmatrix}$

   $\operatorname{im}\left(e^{(1+2i)t}\mathbf{v}\right) = e^t\begin{pmatrix}\sin(2t)\\ -\cos(2t)\\ 0\end{pmatrix}$

   (b) A suitable UC form is

   $\mathbf{y}_p(t) =$ $e^{-2t}\mathbf{a}$

   (c) To determine the UC coefficients you would row reduce

   $\boldsymbol{M}_{\text{UC}} =$ $\left[\begin{array}{c|c} -2\boldsymbol{I}-\boldsymbol{A} & \begin{matrix}-3\\-5\\-1\end{matrix}\end{array}\right]$

   $(-2\boldsymbol{I}-\boldsymbol{A})\mathbf{a} = \begin{pmatrix}-3\\-5\\-1\end{pmatrix}$

   $-2\boldsymbol{I}-\boldsymbol{A} = \begin{pmatrix}-3 & 2 & 0\\ -2 & -3 & 0\\ 0 & 0 & -1\end{pmatrix}$

   $\mathbf{a} = \begin{pmatrix}19\\9\\13\end{pmatrix}/13$

   (d) To satisfy the IC you would row reduce $\boldsymbol{M}_{\text{IC}} =$ $\left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{7}{13}\\ 0 & 1 & 0 & \frac{30}{13}\\ 0 & 0 & 1 & 0\end{array}\right]$

   $\boldsymbol{M}_{\text{IC}} = \left[\,\boldsymbol{F}_1(0)\middle|\mathbf{y}(0)-\mathbf{a}\,\right]$

   $\mathbf{y}(0) - \mathbf{y}_p(0) = \begin{pmatrix}2\\3\\1\end{pmatrix} - \mathbf{a}$

   $= \begin{pmatrix}26-19\\39-9\\13-13\end{pmatrix}/13 = \begin{pmatrix}7\\30\\0\end{pmatrix}/13$

2. $\mathbf{y}' = \boldsymbol{A}\mathbf{y} + \cos(2t)\begin{pmatrix}-3\\-5\\-1\end{pmatrix}$ $\quad \boldsymbol{A} = \begin{pmatrix}-1 & -2 & 1\\ -2 & -3 & 3\\ -1 & -3 & 2\end{pmatrix}$ $\quad \mathbf{y}(0) = \begin{pmatrix}2\\3\\1\end{pmatrix}$

Eigenpairs i, $\begin{pmatrix}1\\-i\\1-i\end{pmatrix}$ $\quad -i$, $\begin{pmatrix}1\\i\\1+i\end{pmatrix}$ $\quad -2$, $\begin{pmatrix}1\\1\\1\end{pmatrix}$

$\alpha = 0, \quad \beta = 1$

$\operatorname{re}\mathbf{v} = \begin{pmatrix}1\\0\\1\end{pmatrix}, \quad \operatorname{im}\mathbf{v} = \begin{pmatrix}0\\-1\\-1\end{pmatrix}$

(a) $\mathbf{y}_c(t) = \boxed{\begin{pmatrix}\cos(t) & -\sin(t) & e^{-2t}\\ \sin(t) & \cos(t) & e^{-2t}\\ \cos(t)+\sin(t) & \cos(t)-\sin(t) & e^{-2t}\end{pmatrix}\mathbf{c}}$

$\operatorname{re}(e^{it}\mathbf{v}) = \begin{pmatrix}\cos(t)\\ \sin(t)\\ \cos(t)+\sin(t)\end{pmatrix}$

$\operatorname{im}(e^{it}\mathbf{v}) = \begin{pmatrix}\sin(t)\\ -\cos(t)\\ \sin(t)-\cos(t)\end{pmatrix}$

(b) A suitable UC form is

$\mathbf{y}_p(t) = \boxed{\mathbf{a}\cos(2t) + \mathbf{b}\sin(2t)}$

(c) To determine the UC coefficients you would row reduce

$$\begin{pmatrix}-\boldsymbol{A} & 2\boldsymbol{I}\\ -2\boldsymbol{I} & -\boldsymbol{A}\end{pmatrix}\begin{pmatrix}\mathbf{a}\\ \mathbf{b}\end{pmatrix} = \begin{pmatrix}\mathbf{f}_0\\ \mathbf{0}\end{pmatrix}$$

$$\boldsymbol{M}_{\text{UC}} = \boxed{\left[\begin{array}{cc|c} & & -3\\ -\boldsymbol{A} & 2\boldsymbol{I} & -5\\ & & -1\\ & & 0\\ -2\boldsymbol{I} & -\boldsymbol{A} & 0\\ & & 0\end{array}\right]}$$

$$\begin{pmatrix}-\boldsymbol{A} & 2\boldsymbol{I}\\ -2\boldsymbol{I} & -\boldsymbol{A}\end{pmatrix} = \begin{pmatrix}1 & 2 & -1 & 2 & 0 & 0\\ 2 & 3 & -3 & 0 & 2 & 0\\ 1 & 3 & -2 & 0 & 0 & 2\\ -2 & 0 & 0 & 1 & 2 & -1\\ 0 & -2 & 0 & 2 & 3 & -3\\ 0 & 0 & -2 & 1 & 3 & -2\end{pmatrix}$$

$$\mathbf{f}_0 = \begin{pmatrix}-3\\-5\\-1\end{pmatrix}$$

$$\mathbf{a} = \begin{pmatrix}-13\\-37\\-29\end{pmatrix}/12, \quad \mathbf{b} = \begin{pmatrix}11\\-5\\27\end{pmatrix}/12$$

(d) To satisfy the IC you would row reduce $\boldsymbol{M}_{\text{IC}} = \boxed{\left[\begin{array}{ccc|c} 1 & 0 & 1 & \frac{37}{12}\\ 0 & 1 & 1 & \frac{73}{12}\\ 1 & 1 & 1 & \frac{41}{12}\end{array}\right]}$

$$\boldsymbol{M}_{\text{IC}} = \left[\, \boldsymbol{F}_2(0) \middle| \mathbf{y}(0) - \mathbf{a} \,\right]$$

$$\mathbf{y}(0) - \mathbf{y}_p(0) = \begin{pmatrix}2\\3\\1\end{pmatrix} - \mathbf{a}$$

$$= \begin{pmatrix}24+13\\ 36+37\\ 12+29\end{pmatrix}/12 = \begin{pmatrix}37\\73\\41\end{pmatrix}/12$$